Cambridge Maths: How to Answer 'Explain the Mistake' Questions

In Stage 6 Cambridge Mathematics exams, many reasoning questions ask you to **spot and explain a mistake** in a student's work. To get full marks, you cannot just write 'it is wrong' or give the correct number—you must clearly explain **WHY** the mistake happened.

Part 1: The 3-Steps to Explain Mistakes

Whenever you see a question asking *"Explain the mistake made in the calculation"*, follow these 3 steps:

Step	Action	What to Write / Examples
Step 1: SPOT IT	Find the exact line or place where the error happens.	Circle the error. Ask yourself: "Which number or sign is wrong?"
Step 2: EXPLAIN IT	Write why the math is incorrect using math vocabulary.	Use sentence starters: "The student forgot to...", "They misaligned...", "They added instead of..."
Step 3: FIX IT	Show the correct working out and write the right answer.	Write down the complete calculation clearly step-by-step.

Word Bank & Sentence Starters

- Useful Verbs: forgot, added, subtracted, multiplied, divided, rounded, confused, placed.
- Key Nouns: place value, decimal point, denominator, numerator, mean, median, mode, outlier, column.
- Sentence Starter 1 (Spotting): "The mistake is in Step [X] where the student..."
- Sentence Starter 2 (Explaining Cause): "They mistakenly [added/forgot to align] because..."
- Sentence Starter 3 (Rule/Generalising): "To calculate [mean/fractions], they should have..."
- Sentence Starter 4 (Correcting): "The correct calculation is... so the correct answer is..."

Topic-by-Topic Worked Examples

Let's look at how to answer 'Explain the Mistake' questions across the three main Stage 6 topic areas.

Topic 1: Addition & Subtraction (Decimals & Place Value)

Student Work Sample (Question): Alex subtracted 3.28 from 15.4 like this:

15.4
- 3.28

12.22 (Alex wrote: 8 - 0 = 8, 4 - 2 = 2, 15 - 3 = 12)

Task: Explain the mistake Alex made and find the correct answer.

File 1 Breakdown (Column Decimal Subtraction)

1. SPOT IT: Look at 15.4 - 3.28. 15.4 has 1 decimal place, but 3.28 has 2 decimal places. Alex did not put a placeholder zero (15.40). Alex just brought down the 8 instead of subtracting 8 from 0!

2. EXPLAIN IT (ESL Model Explanation): "Alex forgot to write a placeholder zero in the hundredths column (15.40). As a result, Alex subtracted 0 from 8 instead of regrouping and subtracting 8 from 10."

3. FIX IT (Correct Math):
   15.40
 - 3.28

   12.12 (Regroup from tenths: 10 - 8 = 2 hundredths, 3 - 2 = 1 tenth, 15 - 3 = 12 units).

Averages (Mean, Mode, Median)

Student Work Sample (Question): Sam calculated the mean of these 4 test scores: 10, 12, 14, 24.
Sam wrote on his calculator: 10 + 12 + 14 + 24 ÷ 4 = 42.
Task: Explain why Sam's answer of 42 is incorrect and find the true mean.

File 2 Breakdown (Mean & Order of Operations)

1. SPOT IT: Look at the scores: 10, 12, 14, 24. The highest score is 24! The mean CANNOT be 42 because the mean must lie between the smallest number (10) and the largest number (24).

2. EXPLAIN IT (ESL Model Explanation): "Sam made an order of operations mistake on his calculator. He divided only the last number (24 ÷ 4 = 6) and added it to 10 + 12 + 14 = 36, giving 42. He forgot to put brackets around the sum before dividing."

3. FIX IT (Correct Math):
   Step 1: Calculate total sum = 10 + 12 + 14 + 24 = 60.
   Step 2: Divide sum by total count = 60 ÷ 4 = 15.
   The correct mean is 15.

Fractions, Decimals & Percentages (FPD)

Student Work Sample (Question): Mia tried to add two fractions: 1/4 + 2/5.
Mia wrote: 1/4 + 2/5 = (1 + 2) / (4 + 5) = 3/9 = 1/3.
Task: Explain the mistake Mia made and calculate the correct answer.

File 3 Breakdown (Adding Fractions with Unlike Denominators)

1. SPOT IT: Mia added the numerators (1 + 2 = 3) AND added the denominators (4 + 5 = 9).

2. EXPLAIN IT (ESL Model Explanation): "Mia mistakenly added the denominators together. You cannot add fractions with different denominators directly. Mia must first convert both fractions to equivalent fractions with a common denominator (20)."

3. FIX IT (Correct Math):
   Common denominator for 4 and 5 is 20.
   1/4 = 5/20 and 2/5 = 8/20
   5/20 + 8/20 = 13/20.
   The correct answer is 13/20.

Student Practice Section ('Spot & Explain the Mistake' Challenges)

Now it is your turn! Read each student sample below. Use the 3-Step Formula and Sentence Starters to write your answer.

Question 1: Addition & Subtraction (Negative Numbers & Temperature)

Ben measured the temperature at 6:00 AM (-4°C) and at 2:00 PM (7°C).
To find the temperature rise, Ben calculated: 7 - 4 = 3°C.
Explain why Ben's answer is wrong and calculate the correct temperature rise.

Explain the mistake here:

_____
_____
_____

Correct Working & Answer:

_____
_____

Question 2: Decimals (Multi-Step Money Word Problem)

Emma bought a book for £6.45 and a pencil for £1.80. She paid with a £20 note.
Emma calculated her change: £20.00 - £6.45 = £13.55. She said her change is £13.55.
Explain the mistake Emma made in solving the word problem.

Explain the mistake here:

_____
_____
_____

Correct Working & Answer:

_____

Question 3: Averages (Selecting the Right Measure of Central Tendency)

A shop owner wants to know the typical shoe size sold in her shop.
Five customers bought shoe sizes: 4, 4, 4, 5, 11.
The shop owner calculated the mean: (4 + 4 + 4 + 5 + 11) ÷ 5 = 28 ÷ 5 = 5.6.
She decided to order shoe size 5.6 for her next stock shipment. Explain why using the mean was a mistake.

Explain the mistake here:

_____
_____
_____

Which average should she use instead and why?:

_____
_____

Question 4: Averages (Working Backwards from a Known Mean)

The mean score of 3 children in a quiz is 8 marks.
A 4th child joins the group and scores 12 marks.
Liam calculated the new mean: (8 + 12) ÷ 2 = 10 marks.
Explain the mistake Liam made in calculating the new mean.

Explain the mistake here:

Correct Working & Answer:

Question 5: Fractions & Decimals (Ordering Mixed Representations)

Lucas was asked to order these values from smallest to largest: 3/5, 0.62, 65%.
Lucas wrote: 3/5, 65%, 0.62 because 3/5 = 0.35, 65% = 0.65, and 0.62.
Explain the two mistakes Lucas made.

Explain the mistakes here:

Correct Working & Order:

Question 6: Percentages (Multi-Step Percentage Discount)

A jacket costs £80. It is on sale with a 25% discount.
Noah calculated the sale price: £80 - 25 = £55.
Explain why Noah's calculation is incorrect and calculate the true sale price.

Explain the mistake here:

Correct Working & Answer:

Part 4: Answer Key & Explanations

Use this section to self-check your answers. Compare your explanation with the model responses to see if you included all key mathematical vocabulary.

Question 1 Answer: Temperature Rise (-4°C to 7°C)

✅ Model Solution & Explanations

• Explanation of Mistake: Ben subtracted 4 from 7 as if the starting temperature was positive 4°C. Because the starting temperature was negative (-4°C), Ben should have calculated the distance from -4°C to 0°C (4 degrees) plus 0°C to 7°C (7 degrees).
• Correct Math: 7 - (-4) = 7 + 4 = 11°C (or 4°C up to 0°C + 7°C up to 7°C = 11°C rise).

Question 2 Answer: Multi-Step Money Problem

✅ Model Solution & Explanations

• Explanation of Mistake: Emma forgot to subtract the cost of the pencil (£1.80). She only subtracted the book (£6.45) from £20.00. In a multi-step problem, she must find the total cost of both items first before subtracting from £20.00.
• Correct Math: Total cost = £6.45 + £1.80 = £8.25. Change = £20.00 - £8.25 = £11.75.

Question 3 Answer: Shoe Size Average Context

✅ Model Solution & Explanations

• Explanation of Mistake: Using the mean was a mistake because shoe sizes are discrete numbers and size 5.6 does not exist. Also, the dataset contains an extreme outlier (size 11) which pulls the mean up away from the typical size.
• Recommended Average: The shop owner should use the Mode (size 4) because it is the most frequent shoe size sold, or the Median (size 4).

Question 4 Answer: Working Backwards from Mean

✅ Model Solution & Explanations

• Explanation of Mistake: Liam averaged the old mean (8) and the new score (12) directly. You cannot average a mean and a single score. Liam forgot to find the original total score of the 3 children first.
• Correct Math: Original total = 3 × 8 = 24 marks. New total with 4th child = 24 + 12 = 36 marks. New mean = 36 ÷ 4 = 9 marks.

Question 5 Answer: Ordering Mixed FPD Values

✅ Model Solution & Explanations

• Explanation of Mistake: Lucas made two errors. First, 3/5 is equal to 0.60 (or 60%), NOT 0.35. Second, he listed 0.62 after 65% (0.65) even though 0.62 is smaller than 0.65.
• Correct Conversion & Order: 3/5 = 60% (0.60), 0.62 = 62%, 65% = 65%. Correct order from smallest to

largest: 3/5, 0.62, 65%.

Question 6 Answer: Percentage Discount

✅ Model Solution & Explanations

• Explanation of Mistake: Noah subtracted the number 25 directly from £80 (£80 - 25 = £55). 25% means 25 out of 100 (or 1/4 of the total amount), NOT £25.
• Correct Math: 25% of £80 = £80 ÷ 4 = £20. Discounted sale price = £80 - £20 = £60.